7.CCNP Enterprise作业七——EIGRP 综合实验

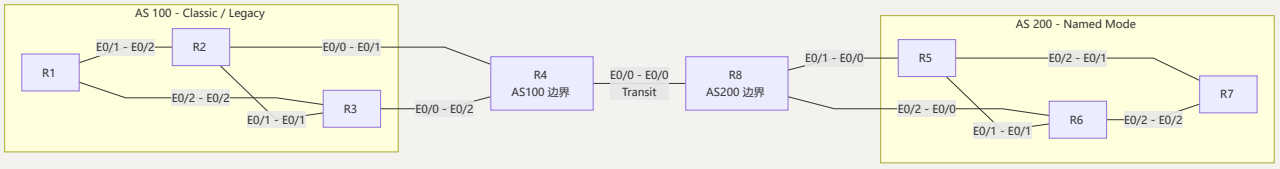
作业主题

EIGRP 双栈综合实验：Classic/Legacy、Named Mode、认证、默认路由、汇总与非等价负载分担

作业目标

1. 掌握 IPv4 EIGRP Classic Mode 的基础配置与验证方法
2. 掌握 IPv6 EIGRP Legacy Mode 的基础配置与验证方法
3. 掌握 EIGRP Named Mode 中 IPv4/IPv6 Address Family 的配置结构
4. 掌握 EIGRP 认证、被动接口、Router ID、手工汇总和默认路由引入
5. 理解两个 EIGRP 域之间只通过默认路由互通、不互相引入明细路由的设计
6. 理解 `maximum-paths`、`variance` 与可行条件对 非等价负载分担的影响
7. 能使用 show 命令验证邻接、拓扑表、路由表、汇总路由和故障收敛

实验拓扑



R4 只属于 AS100，R8 只属于 AS200。R4-R8 中间链路只作为普通三层 Transit 链路，不运行任何 EIGRP。

接口及 **IP** 地址规划

设备	接口	类型	**IPV4** 规划	**IPV6** 规划
R1	E0/1	AS100	`10.1.12.1/24`	`2001:db8:12::1/64`
R1	E0/2	AS100	`10.1.13.1/24`	`2001:db8:13::1/64`
R2	E0/2	AS100	`10.1.12.2/24`	`2001:db8:12::2/64`
R2	E0/1	AS100	`10.1.23.2/24`	`2001:db8:23::2/64`
R2	E0/0	AS100	`10.1.24.2/24`	`2001:db8:24::2/64`
R3	E0/2	AS100	`10.1.13.3/24`	`2001:db8:13::3/64`
R3	E0/1	AS100	`10.1.23.3/24`	`2001:db8:23::3/64`
R3	E0/0	AS100	`10.1.34.3/24`	`2001:db8:34::3/64`
R4	E0/1	AS100	`10.1.24.4/24`	`2001:db8:24::4/64`
R4	E0/2	AS100	`10.1.34.4/24`	`2001:db8:34::4/64`
R4	E0/0	Transit	`10.1.48.4/24`	`2001:db8:48::4/64`
R8	E0/0	Transit	`10.1.48.8/24`	`2001:db8:48::8/64`
R8	E0/1	AS200	`10.1.58.8/24`	`2001:db8:58::8/64`
R8	E0/2	AS200	`10.1.68.8/24`	`2001:db8:68::8/64`
R5	E0/0	AS200	`10.1.58.5/24`	`2001:db8:58::5/64`
R5	E0/1	AS200	`10.1.56.5/24`	`2001:db8:56::5/64`
R5	E0/2	AS200	`10.1.57.5/24`	`2001:db8:57::5/64`
R6	E0/0	AS200	`10.1.68.6/24`	`2001:db8:68::6/64`
R6	E0/1	AS200	`10.1.56.6/24`	`2001:db8:56::6/64`
R6	E0/2	AS200	`10.1.67.6/24`	`2001:db8:67::6/64`
R7	E0/1	AS200	`10.1.57.7/24`	`2001:db8:57::7/64`
R7	E0/2	AS200	`10.1.67.7/24`	`2001:db8:67::7/64`

Loopback 地址规划

设备	**LOOPBACK**	类型	**IPV4** 规划	**IPV6** 规划
R1	Loopback0	Router ID	`1.1.1.1/32`	`2001:db8:1:ffff::1/128`

设备	**LOOPBACK**	类型	**IPV4** 规划	**IPV6** 规划
R1	Loopback10	业务网段	`10.100.1.1/24`	`2001:db8:100:1::1/64`
R2	Loopback0	Router ID	`2.2.2.2/32`	`2001:db8:2:ffff::2/128`
R2	Loopback10	业务网段	`10.100.2.2/24`	`2001:db8:100:2::2/64`
R3	Loopback0	Router ID	`3.3.3.3/32`	`2001:db8:3:ffff::3/128`
R3	Loopback10	业务网段	`10.100.3.3/24`	`2001:db8:100:3::3/64`
R4	Loopback0	Router ID	`4.4.4.4/32`	`2001:db8:4:ffff::4/128`
R8	Loopback0	Router ID	`8.8.8.8/32`	`2001:db8:8:ffff::8/128`
R5	Loopback0	Router ID	`5.5.5.5/32`	`2001:db8:5:ffff::5/128`
R5	Loopback10	业务网段	`10.200.5.5/24`	`2001:db8:200:5::5/64`
R6	Loopback0	Router ID	`6.6.6.6/32`	`2001:db8:6:ffff::6/128`
R6	Loopback10	业务网段	`10.200.6.6/24`	`2001:db8:200:6::6/64`
R7	Loopback0	Router ID	`7.7.7.7/32`	`2001:db8:7:ffff::7/128`
R7	Loopback70	业务网段	`10.200.70.7/24`	`2001:db8:200:70::7/64`
R7	Loopback71	业务网段	`10.200.71.7/24`	`2001:db8:200:71::7/64`

接口连接规划

设备	接口	连接设备	连接接口
R1	E0/1	R2	E0/2
R1	E0/2	R3	E0/2
R2	E0/1	R3	E0/1
R2	E0/0	R4	E0/1
R3	E0/0	R4	E0/2
R4	E0/0	R8	E0/0
R8	E0/1	R5	E0/0
R8	E0/2	R6	E0/0
R5	E0/1	R6	E0/1
R5	E0/2	R7	E0/1
R6	E0/2	R7	E0/2

EIGRP 域规划

EIGRP 域	设备	**IPV4** 模式	**IPV6** 模式	说明
AS100	R1、R2、R3、R4	Classic Mode	Legacy EIGRP	R4 为 AS100 边界
AS200	R8、R5、R6、R7	Named Mode	Named Mode	R8 为 AS200 边界
Transit	R4、R8	不运行 EIGRP	不运行 EIGRP	只配置三层地址

认证规划

位置	地址族	认证方式	**KEY-CHAIN** / 密钥
AS100 所有 EIGRP 邻接接口	IPv4	MD5	EIGRP-AS100-MD5 / CcnpAs100-2026
AS100 所有 EIGRP 邻接接口	IPv6	MD5	EIGRP-AS100-MD5 / CcnpAs100-2026
AS200 所有 EIGRP 邻接接口	IPv4	MD5	EIGRP-AS200-MD5 / CcnpAs200-2026
AS200 所有 EIGRP 邻接接口	IPv6	HMAC-SHA-256	CcnpAs200Sha-2026

默认路由规划

设备	静态默认路由	引入位置	种子 **METRIC**
R4	`0.0.0.0/0` 指向 `10.1.48.8`	EIGRP 100 IPv4	`10000 100 255 1 1500`
R4	`::/0` 指向 `2001:db8:48::8`	EIGRP 100 IPv6	`10000 100 255 1 1500`
R8	`0.0.0.0/0` 指向 `10.1.48.4`	AS200 IPv4 AF	`10000 100 255 1 1500`
R8	`::/0` 指向 `2001:db8:48::4`	AS200 IPv6 AF	`10000 100 255 1 1500`

本实验不允许配置 `redistribute eigrp 100` 或 `redistribute eigrp 200`，也不需要使用 prefix-list、route-map 或 distribute-list。R4、R8 上只引入本实验指定的静态默认路由。

汇总与负载分担规划

项目	配置位置	要求
IPv4 汇总	R7 的 E0/1、E0/2	向 R5、R6 发布 `10.200.0.0/16`
IPv6 汇总	R7 的 E0/1、E0/2	向 R5、R6 发布 `2001:db8:200::/48`
非等价负载分担	AS200	`maximum-paths 4`，`variance 2`
Metric 调整	R8 的 E0/2	只配置 `delay 200`，其他以太网接口保持默认 Metric

配置需求

1. 根据实验拓扑完成设备连接与基础配置
2. 在所有路由器上开启 IPv6 单播转发
3. 按照规划表完成所有物理接口、Loopback 接口的 IPv4/IPv6 地址配置
4. 在 R1、R2、R3、R4 上配置 AS100 的 IPv4 EIGRP Classic Mode
5. 在 R1、R2、R3、R4 上配置 AS100 的 IPv6 EIGRP Legacy Mode
6. 在 AS100 中配置 Router ID、被动接口和传统接口认证
7. 在 R8、R5、R6、R7 上配置 AS200 的 EIGRP Named Mode
8. 在 AS200 中分别配置 IPv4 Address Family 与 IPv6 Address Family
9. 在 AS200 中配置 Router ID、被动接口、IPv4 MD5 认证和 IPv6 HMAC-SHA-256 认证
10. 确保 R4-R8 的 E0/0 Transit 链路不加入任何 EIGRP AS
11. 在 R4 上配置静态默认路由指向 R8，并将静态默认路由引入 AS100
12. 在 R8 上配置静态默认路由指向 R4，并将静态默认路由引入 AS200
13. 在 R7 面向 R5、R6 的接口上配置 IPv4/IPv6 手工汇总
14. 在 R8 的 E0/2 上配置 `delay 200`，用于观察非等价负载分担
15. 验证 AS100 与 AS200 内部邻接、拓扑表和路由表是否正常
16. 验证 AS100 与 AS200 只通过默认路由跨域通信，不互相学习对方业务明细路由
17. 验证关闭指定链路后的 EIGRP 收敛现象

验证要求

1. 使用 `show ip eigrp neighbors` 和 `show ipv6 eigrp neighbors` 验证 AS100 邻接
2. 使用 `show eigrp address-family ipv4 neighbors` 和 `show eigrp address-family ipv6 neighbors` 验证 AS200 邻接
3. 使用 `show ip protocols` 或 `show eigrp protocols` 验证 Router ID、Network、Passive、Maximum Paths 和 Variance
4. 在 R1、R7 上验证是否学习到 EIGRP 默认路由
5. 在 R1 上验证不应学习到 AS200 的 `10.200.0.0/16` 或其他 AS200 明细为 `D EX`
6. 在 R7 上验证不应学习到 AS100 的 `10.100.0.0/16` 或其他 AS100 明细为 `D EX`
7. 从 R1 的业务 Loopback ping R7 的业务 Loopback，验证 IPv4/IPv6 跨域连通性
8. 在 R8 上查看到 R7 汇总网段的拓扑信息，观察多候选路径与 `variance 2` 的效果
9. 关闭 R8 到 R5 的链路，观察 AS200 内部重新收敛
10. 关闭 R4 到 R2 的链路，观察 AS100 内部重新收敛
11. 关闭 R4-R8 Transit 链路，验证该链路故障不会导致 EIGRP 邻接变化，但会影响跨域业务通信

作业要求

1. 请提交 R1、R2、R3、R4、R5、R6、R7、R8 的最终配置

```
!
! Last configuration change at 08:50:42 UTC Wed Nov 4 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
```

```
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!


!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
```

```
key chain EIGRP-AS100-MD5
 key 1
  key-string CcnpAs100-2026
!
!
!
!
!
!
!
!
redundancy
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Loopback0
 no shutdown
 ip address 1.1.1.1 255.255.255.255
 ipv6 address 2001:DB8:1:FFFF::1/128
 ipv6 eigrp 100
!
interface Loopback10
 no shutdown
 ip address 10.100.1.1 255.255.255.0
 ipv6 address 2001:DB8:100:1::1/64
 ipv6 eigrp 100
!
interface Ethernet0/0
```

```
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet0/1
 no shutdown
 ip address 10.1.12.1 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:12::1/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/2
 no shutdown
 ip address 10.1.13.1 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:13::1/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/1
```

```
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/3
 no shutdown
 no ip address
 shutdown
 duplex auto
```

```
!
interface Ethernet3/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
!
router eigrp 100
 network 1.1.1.1 0.0.0.0
 network 10.1.12.0 0.0.0.255
 network 10.1.13.0 0.0.0.255
 network 10.100.1.0 0.0.0.255
 passive-interface default
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 eigrp router-id 1.1.1.1
!
ip forward-protocol nd
!
!
no ip http server
```

```
no ip http secure-server
!
ipv6 router eigrp 100
 passive-interface default
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 no passive-interface Loopback0
 no passive-interface Loopback10
 eigrp router-id 1.1.1.1
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input none
!
!
end
```

r2

```
!
! Last configuration change at 08:41:19 UTC Wed Nov 4 2026
!
```

```
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R2
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!


!
!
!
```

```
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
key chain EIGRP-AS100-MD5
 key 1
  key-string CcnpAs100-2026
!
!
!
!
!
!
!
!
redundancy
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Loopback0
 no shutdown
 ip address 2.2.2.2 255.255.255.255
```

```
 ipv6 address 2001:DB8:2:FFFF::2/128
 ipv6 eigrp 100
!
interface Loopback10
 no shutdown
 ip address 10.100.2.2 255.255.255.0
 ipv6 address 2001:DB8:100:2::2/64
 ipv6 eigrp 100
!
interface Ethernet0/0
 no shutdown
 ip address 10.1.24.2 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:24::2/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/1
 no shutdown
 ip address 10.1.23.2 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:23::2/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/2
 no shutdown
 ip address 10.1.12.2 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:12::2/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
```

```
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/1
 no shutdown
```

```
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
```

```
!
router eigrp 100
 network 2.2.2.2 0.0.0.0
 network 10.1.12.0 0.0.0.255
 network 10.1.23.0 0.0.0.255
 network 10.1.24.0 0.0.0.255
 network 10.100.2.0 0.0.0.255
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 eigrp router-id 2.2.2.2
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 router eigrp 100
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 no passive-interface Loopback0
 no passive-interface Loopback10
 eigrp router-id 2.2.2.2
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
```

```
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input none
!
!
end
```

r3

```
!
! Last configuration change at 08:44:12 UTC Wed Nov 4 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R3
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
```

```
!
!
!
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!


!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
key chain EIGRP-AS100-MD5
 key 1
  key-string CcnpAs100-2026
!
!
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!
!
!
!
```

```
redundancy
!
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!
!
!
!
interface Loopback0
 no shutdown
 ip address 3.3.3.3 255.255.255.255
 ipv6 address 2001:DB8:3:FFFF::3/128
 ipv6 eigrp 100
!
interface Loopback10
 no shutdown
 ip address 10.100.3.3 255.255.255.0
 ipv6 address 2001:DB8:100:3::3/64
 ipv6 eigrp 100
!
interface Ethernet0/0
 no shutdown
 ip address 10.1.34.3 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:34::3/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/1
```

```
 no shutdown
 ip address 10.1.23.3 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:23::3/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/2
 no shutdown
 ip address 10.1.13.3 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:13::3/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/2
```

```
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/0
 no shutdown
 no ip address
 shutdown
 duplex auto
```

```
!
interface Ethernet3/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
!
router eigrp 100
 network 3.3.3.3 0.0.0.0
 network 10.1.13.0 0.0.0.255
 network 10.1.23.0 0.0.0.255
 network 10.1.34.0 0.0.0.255
 network 10.100.3.0 0.0.0.255
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 eigrp router-id 3.3.3.3
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 router eigrp 100
 passive-interface default
```

```
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 eigrp router-id 3.3.3.3
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input none
!
!
end
```

r4

```
!
! Last configuration change at 13:00:04 UTC Wed Nov 4 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
```

```
!
hostname R4
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!


!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
```

```
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
key chain EIGRP-AS100-MD5
 key 1
  key-string CcnpAs100-2026
!
!
!
!
!
!
!
!
!
redundancy
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Loopback0
 no shutdown
 ip address 4.4.4.4 255.255.255.255
 ipv6 address 2001:DB8:4:FFFF::4/128
 ipv6 eigrp 100
!
interface Ethernet0/0
```

```
 no shutdown
 ip address 10.1.48.4 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:48::4/64
!
interface Ethernet0/1
 no shutdown
 ip address 10.1.24.4 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:24::4/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/2
 no shutdown
 ip address 10.1.34.4 255.255.255.0
 ip authentication mode eigrp 100 md5
 ip authentication key-chain eigrp 100 EIGRP-AS100-MD5
 duplex auto
 ipv6 address 2001:DB8:34::4/64
 ipv6 eigrp 100
 ipv6 authentication mode eigrp 100 md5
 ipv6 authentication key-chain eigrp 100 EIGRP-AS100-MD5
!
interface Ethernet0/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/1
```

```
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/3
 no shutdown
 no ip address
 shutdown
 duplex auto
```

```
!
interface Ethernet3/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
!
router eigrp 100
 network 4.4.4.4 0.0.0.0
 network 10.1.24.0 0.0.0.255
 network 10.1.34.0 0.0.0.255
 redistribute static metric 10000 100 255 1 1500
 passive-interface default
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 eigrp router-id 4.4.4.4
!
ip forward-protocol nd
!
!
no ip http server
```

```
no ip http secure-server
ip route 0.0.0.0 0.0.0.0 10.1.48.8
!
ipv6 route ::/0 2001:DB8:48::8
ipv6 router eigrp 100
 passive-interface default
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 eigrp router-id 4.4.4.4
 redistribute static metric 10000 100 255 1 1500
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input none
!
!
end
```

r5

```

```

```
!
! Last configuration change at 12:12:06 UTC Wed Nov 4 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R5
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
key chain EIGRP-AS200-MD5
 key 1
  key-string CcnpAs200-2026
!
!
!
!
!
!
!
!
!
redundancy
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
```

```
interface Loopback0
 no shutdown
 ip address 5.5.5.5 255.255.255.255
 ipv6 address 2001:DB8:5:FFFF::5/128
!
interface Loopback10
 no shutdown
 ip address 10.200.5.5 255.255.255.0
 ipv6 address 2001:DB8:200:5::5/64
!
interface Ethernet0/0
 no shutdown
 ip address 10.1.58.5 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:58::5/64
!
interface Ethernet0/1
 no shutdown
 ip address 10.1.56.5 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:56::5/64
!
interface Ethernet0/2
 no shutdown
 ip address 10.1.57.5 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:57::5/64
!
interface Ethernet0/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
```

```
interface Ethernet1/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/3
 no shutdown
 no ip address
 shutdown
```

```
 duplex auto
!
interface Ethernet3/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
!
router eigrp AS200
 !
 address-family ipv4 unicast autonomous-system 200
  !
  af-interface default
   authentication mode md5
   authentication key-chain EIGRP-AS200-MD5
   passive-interface
  exit-af-interface
  !
  af-interface Ethernet0/1
   no passive-interface
  exit-af-interface
```

```
 !
 af-interface Ethernet0/0
  no passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/2
  no passive-interface
 exit-af-interface
 !
 topology base
  variance 2
 exit-af-topology
 network 5.5.5.5 0.0.0.0
 network 10.1.56.0 0.0.0.255
 network 10.1.57.0 0.0.0.255
 network 10.1.58.0 0.0.0.255
 network 10.200.5.0 0.0.0.255
 eigrp router-id 5.5.5.5
exit-address-family
!
address-family ipv6 unicast autonomous-system 200
 !
 af-interface default
  authentication mode hmac-sha-256 CcnpAs200Sha-2026
  passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/0
  no passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/1
  no passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/2
  no passive-interface
 exit-af-interface
 !
 topology base
```

```
   maximum-paths 4
   variance 2
  exit-af-topology
  eigrp router-id 5.5.5.5
 exit-address-family
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input none
!
!
end
```

r6

```
```

```
!
! Last configuration change at 12:02:54 UTC Wed Nov 4 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R6
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
key chain EIGRP-AS200-MD5
 key 1
  key-string CcnpAs200-2026
!
!
!
!
!
!
!
!
!
redundancy
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
```

```
interface Loopback0
 no shutdown
 ip address 6.6.6.6 255.255.255.255
 ipv6 address 2001:DB8:6:FFFF::6/128
!
interface Loopback10
 no shutdown
 ip address 10.200.6.6 255.255.255.0
 ipv6 address 2001:DB8:200:6::6/64
!
interface Ethernet0/0
 no shutdown
 ip address 10.1.68.6 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:68::6/64
!
interface Ethernet0/1
 no shutdown
 ip address 10.1.56.6 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:56::6/64
!
interface Ethernet0/2
 no shutdown
 ip address 10.1.67.6 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:67::6/64
!
interface Ethernet0/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
```

```
interface Ethernet1/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/3
 no shutdown
 no ip address
 shutdown
```

```
 duplex auto
!
interface Ethernet3/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
!
router eigrp AS200
 !
 address-family ipv4 unicast autonomous-system 200
  !
  af-interface default
   authentication mode md5
   authentication key-chain EIGRP-AS200-MD5
   passive-interface
  exit-af-interface
  !
  af-interface Ethernet0/0
   no passive-interface
  exit-af-interface
```

```
 !
 af-interface Ethernet0/1
  no passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/2
  no passive-interface
 exit-af-interface
 !
 topology base
  variance 2
 exit-af-topology
 network 6.6.6.6 0.0.0.0
 network 10.1.56.0 0.0.0.255
 network 10.1.67.0 0.0.0.255
 network 10.1.68.0 0.0.0.255
 network 10.200.6.0 0.0.0.255
 eigrp router-id 6.6.6.6
exit-address-family
!
address-family ipv6 unicast autonomous-system 200
 !
 af-interface default
  authentication mode hmac-sha-256 CcnpAs200Sha-2026
  passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/0
  no passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/1
  no passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/2
  no passive-interface
 exit-af-interface
 !
 topology base
```

```
   maximum-paths 4
   variance 2
  exit-af-topology
  eigrp router-id 6.6.6.6
 exit-address-family
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input none
!
!
end
```

r7

```
```

```
!
! Last configuration change at 12:48:30 UTC Wed Nov 4 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R7
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
```

```
!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
key chain EIGRP-AS200-MD5
 key 1
  key-string CcnpAs200-2026
!
!
!
!
!
!
!
!
!
redundancy
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
```

```
interface Loopback0
 no shutdown
 ip address 7.7.7.7 255.255.255.255
 ipv6 address 2001:DB8:7:FFFF::7/128
!
interface Loopback70
 no shutdown
 ip address 10.200.70.7 255.255.255.0
 ipv6 address 2001:DB8:200:70::7/64
!
interface Loopback71
 no shutdown
 ip address 10.200.71.7 255.255.255.0
 ipv6 address 2001:DB8:200:71::7/64
!
interface Ethernet0/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet0/1
 no shutdown
 ip address 10.1.57.7 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:57::7/64
!
interface Ethernet0/2
 no shutdown
 ip address 10.1.67.7 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:67::7/64
!
interface Ethernet0/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/0
```

```
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/2
 no shutdown
 no ip address
 shutdown
 duplex auto
```

```
!
interface Ethernet2/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
!
router eigrp AS200
 !
 address-family ipv4 unicast autonomous-system 200
  !
  af-interface default
   authentication mode md5
   authentication key-chain EIGRP-AS200-MD5
   passive-interface
```

```
 exit-af-interface
 !
 af-interface Ethernet0/1
  summary-address 10.200.0.0 255.255.0.0
  no passive-interface
 exit-af-interface
 !
 af-interface Ethernet0/2
  summary-address 10.200.0.0 255.255.0.0
  no passive-interface
 exit-af-interface
 !
 topology base
  variance 2
 exit-af-topology
 network 7.7.7.7 0.0.0.0
 network 10.1.57.0 0.0.0.255
 network 10.1.67.0 0.0.0.255
 network 10.200.70.0 0.0.0.255
 network 10.200.71.0 0.0.0.255
 eigrp router-id 7.7.7.7
exit-address-family
!
address-family ipv6 unicast autonomous-system 200
 !
 af-interface default
  shutdown
 exit-af-interface
 !
 af-interface Ethernet0/1
  summary-address 2001:DB8:200::/48
  no shutdown
  authentication mode hmac-sha-256 CcnpAs200Sha-2026
 exit-af-interface
 !
 af-interface Ethernet0/0
  authentication mode hmac-sha-256 CcnpAs200Sha-2026
 exit-af-interface
 !
 af-interface Ethernet0/2
```

```
   summary-address 2001:DB8:200::/48
   no shutdown
   authentication mode hmac-sha-256 CcnpAs200Sha-2026
  exit-af-interface
  !
  af-interface Loopback0
   no shutdown
   passive-interface
  exit-af-interface
  !
  af-interface Loopback70
   no shutdown
   passive-interface
  exit-af-interface
  !
  af-interface Loopback71
   no shutdown
   passive-interface
  exit-af-interface
  !
  topology base
   maximum-paths 4
   variance 2
  exit-af-topology
  eigrp router-id 7.7.7.7
 exit-address-family
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
ipv6 ioam timestamp
!
!
!
control-plane
!
!
```

```
!
!
!
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input none
!
!
end
```

r8

```
!
! Last configuration change at 13:00:44 UTC Wed Nov 4 2026
!
version 15.7
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R8
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
```

```
mmi polling-interval 60
no mmi auto-configure
no mmi pvc
mmi snmp-timeout 180
!
!
!
!
!
!
!
!
!
!
!
!
!
!


!
!
!
!
no ip domain lookup
ip cef
ipv6 unicast-routing
ipv6 cef
!
multilink bundle-name authenticated
!
!
!
key chain EIGRP-AS200-MD5
 key 1
  key-string CcnpAs200-2026
!
!
!
```

```
!
!
!
!
!
redundancy
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Loopback0
 no shutdown
 ip address 8.8.8.8 255.255.255.255
 ipv6 address 2001:DB8:8:FFFF::8/128
!
interface Ethernet0/0
 no shutdown
 ip address 10.1.48.8 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:48::8/64
!
interface Ethernet0/1
 no shutdown
 ip address 10.1.58.8 255.255.255.0
 duplex auto
 ipv6 address 2001:DB8:58::8/64
!
interface Ethernet0/2
 no shutdown
```

```
 ip address 10.1.68.8 255.255.255.0
 delay 200
 duplex auto
 ipv6 address 2001:DB8:68::8/64
!
interface Ethernet0/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet1/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/0
 no shutdown
 no ip address
 shutdown
 duplex auto
```

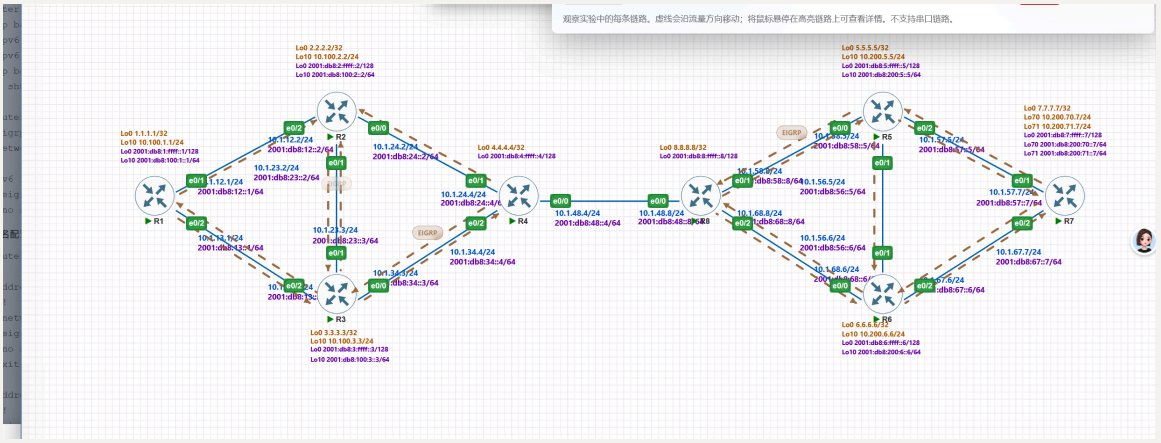
```
!
interface Ethernet2/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet2/3
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/0
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/1
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/2
 no shutdown
 no ip address
 shutdown
 duplex auto
!
interface Ethernet3/3
 no shutdown
 no ip address
```

```
 shutdown
 duplex auto
!
!
router eigrp AS200
 !
 address-family ipv4 unicast autonomous-system 200
  !
  af-interface default
   authentication mode md5
   authentication key-chain EIGRP-AS200-MD5
   passive-interface
  exit-af-interface
  !
  af-interface Ethernet0/1
   no passive-interface
  exit-af-interface
  !
  af-interface Ethernet0/2
   no passive-interface
  exit-af-interface
  !
  topology base
   variance 2
   redistribute static metric 10000 100 255 1 1500
  exit-af-topology
  network 8.8.8.8 0.0.0.0
  network 10.1.58.0 0.0.0.255
  network 10.1.68.0 0.0.0.255
  eigrp router-id 8.8.8.8
 exit-address-family
 !
 address-family ipv6 unicast autonomous-system 200
  !
  af-interface default
   shutdown
   authentication mode hmac-sha-256 CcnpAs200Sha-2026
   passive-interface
  exit-af-interface
  !
```

```
  af-interface Ethernet0/1
   no shutdown
   no passive-interface
  exit-af-interface
  !
  af-interface Ethernet0/2
   no shutdown
   no passive-interface
  exit-af-interface
  !
  af-interface Loopback0
   no shutdown
  exit-af-interface
  !
  topology base
   maximum-paths 4
   variance 2
   redistribute static metric 10000 100 255 1 1500
  exit-af-topology
  eigrp router-id 8.8.8.8
 exit-address-family
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 0.0.0.0 0.0.0.0 10.1.48.4
!
ipv6 route ::/0 2001:DB8:48::4
ipv6 ioam timestamp
!
!
!
control-plane
!
!
!
!
!
```

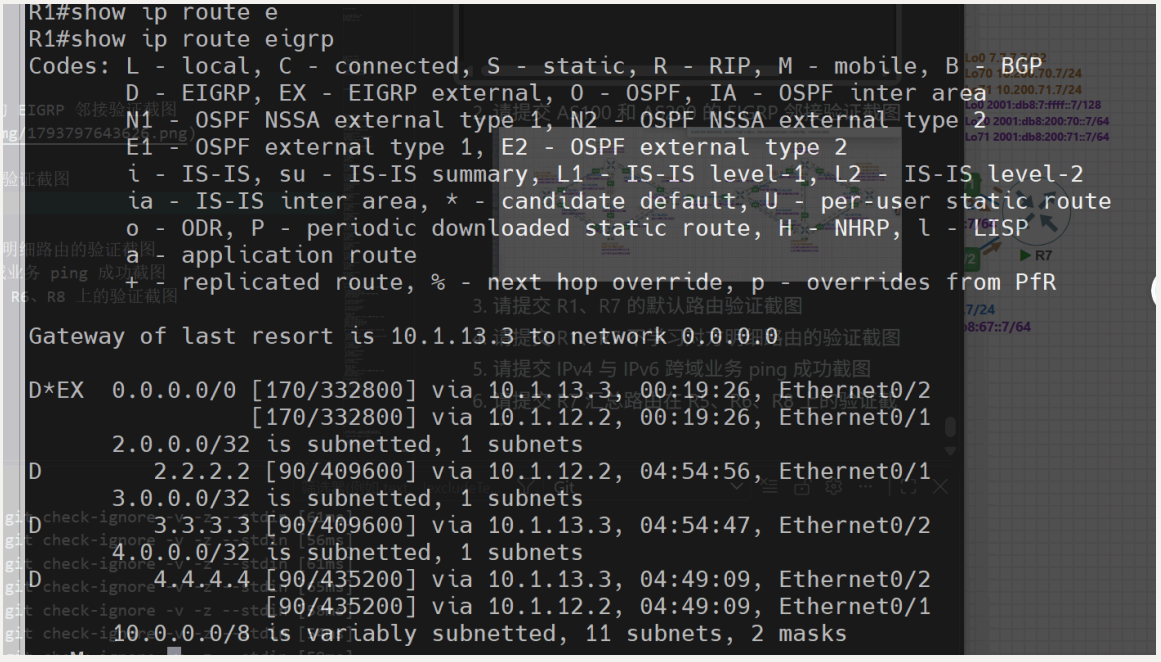
```
!
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
 transport input none
!
!
end
```

2. 请提交 AS100 和 AS200 的 EIGRP 邻接验证截图



3. 请提交 R1、R7 的默认路由验证截图

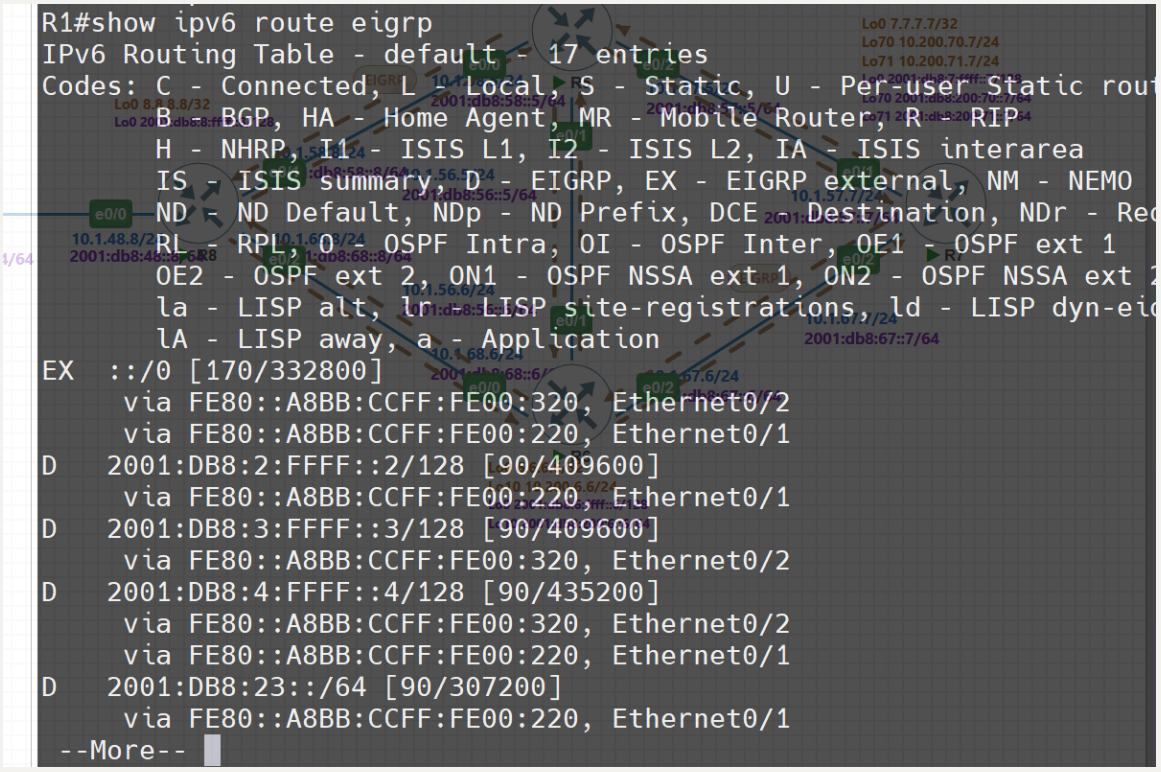
R1



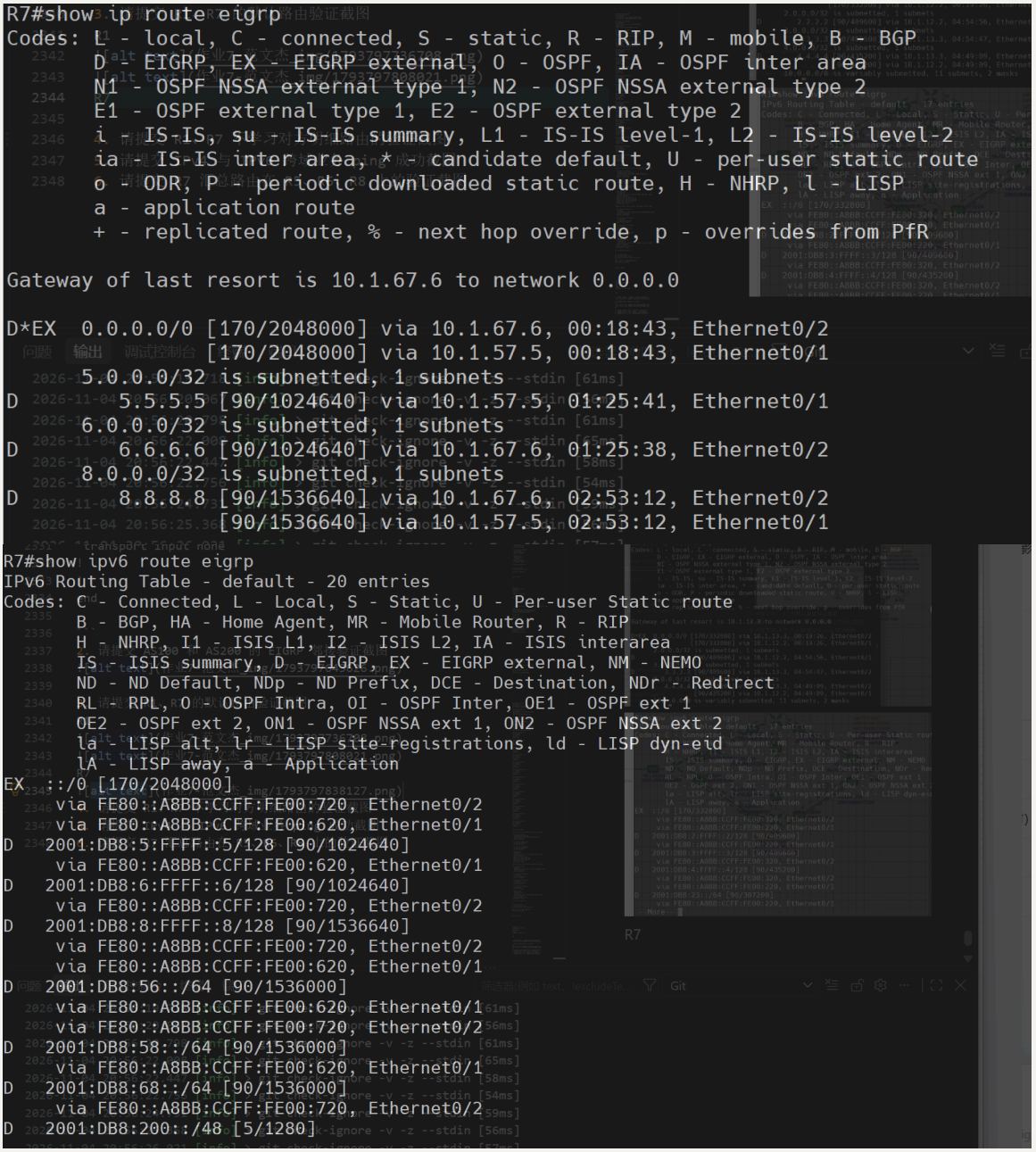
```
R1#show ip route e
R1#show ip route eigrp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.1.13.3 to network 0.0.0.0

D*EX  0.0.0.0/0 [170/332800] via 10.1.13.3, 00:19:26, Ethernet0/2
                [170/332800] via 10.1.12.2, 00:19:26, Ethernet0/1
      2.0.0.0/32 is subnetted, 1 subnets
D        2.2.2.2 [90/409600] via 10.1.12.2, 04:54:56, Ethernet0/1
      3.0.0.0/32 is subnetted, 1 subnets
D        3.3.3.3 [90/409600] via 10.1.13.3, 04:54:47, Ethernet0/2
      4.0.0.0/32 is subnetted, 1 subnets
D        4.4.4.4 [90/435200] via 10.1.13.3, 04:49:09, Ethernet0/2
                [90/435200] via 10.1.12.2, 04:49:09, Ethernet0/1
      10.0.0.0/8 is variably subnetted, 11 subnets, 2 masks
```

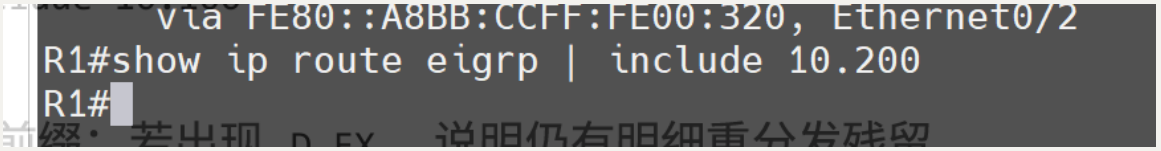

R1#show ipv6 route eigrp
IPv6 Routing Table - default - 17 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static rou
B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Rec
RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
lA - LISP away, a - Application
EX ::/0 [170/332800]
via FE80::A8BB:CCFF:FE00:320, Ethernet0/2
via FE80::A8BB:CCFF:FE00:220, Ethernet0/1
D 2001:DB8:2:FFFF::2/128 [90/409600]
via FE80::A8BB:CCFF:FE00:220, Ethernet0/1
D 2001:DB8:3:FFFF::3/128 [90/409600]
via FE80::A8BB:CCFF:FE00:320, Ethernet0/2
D 2001:DB8:4:FFFF::4/128 [90/435200]
via FE80::A8BB:CCFF:FE00:320, Ethernet0/2
via FE80::A8BB:CCFF:FE00:220, Ethernet0/1
D 2001:DB8:23::/64 [90/307200]
via FE80::A8BB:CCFF:FE00:220, Ethernet0/1
--More--

R7

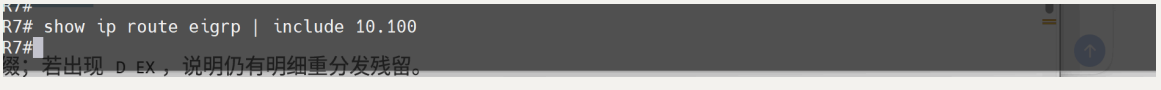


4. 请提交 R1、R7 不学习对方明细路由的验证截图

r1:



r7



5. 请提交 IPv4 与 IPv6 跨域业务 ping 成功截图

ipv4

```
R1#show ip route eigrp | include 10.200
R1# ping 10.200.70.7 source 10.100.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.200.70.7, timeout is 2 seconds:
Packet sent with a source address of 10.100.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#
R7#ping 10.100.1.1 source 10.200.70.7
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.100.1.1, timeout is 2 seconds:
Packet sent with a source address of 10.200.70.7
```

ipv6:

```
R1#ping 2001:db8:200:70::7 source 2001:db8:100:1::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:200:70::7, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:100:1::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#
R1#
R1#Packet sent with a source address of 2001:DB8:100:1::1
      ^
% Invalid input detected at '^' marker.

R1#show ipv6 route eigrp | include 2001:DB8:200
R1#
```

```
R7#show ipv6 route eigrp | include 2001:DB8:100
R7#ping 2001:db8:100:1::1 source 2001:db8:200:70::7
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:100:1::1, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:200:70::7
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R7#
```

6. 请提交 R7 汇总路由在 R5、R6、R8 上的验证截图

r5

```
R5#show ip route 10.200.0.0
Routing entry for 10.200.0.0/16
  Known via "eigrp 200", distance 90, metric 1024640, type internal
  Redistributing via eigrp 200
  Last update from 10.1.57.7 on Ethernet0/2, 00:41:10 ago
  Routing Descriptor Blocks:
  * 10.1.57.7, from 10.1.57.7, 00:41:10 ago, via Ethernet0/2
      Route metric is 1024640, traffic share count is 1
      Total delay is 1002 microseconds, minimum bandwidth is 10000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 1
R5#
```

r6:

```
R6#show ip route 10.200.0.0
Routing entry for 10.200.0.0/16
  Known via "eigrp 200", distance 90, metric 1024640, type internal
  Redistributing via eigrp 200
  Last update from 10.1.67.7 on Ethernet0/2, 00:41:41 ago
  Routing Descriptor Blocks:
  * 10.1.67.7, from 10.1.67.7, 00:41:41 ago, via Ethernet0/2
      Route metric is 1024640, traffic share count is 1
      Total delay is 1002 microseconds, minimum bandwidth is 10000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 1
R6#
```

r8:

```
R8>enable
R8#
R8#show ip route 10.200.0.0
Routing entry for 10.200.0.0/16
  Known via "eigrp 200", distance 90, metric 1536640, type internal
  Redistributing via eigrp 200
  Last update from 10.1.68.6 on Ethernet0/2, 00:42:19 ago
  Routing Descriptor Blocks:
    10.1.68.6, from 10.1.68.6, 00:42:19 ago, via Ethernet0/2
      Route metric is 2048640, traffic share count is 3
      Total delay is 3002 microseconds, minimum bandwidth is 10000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 2
  * 10.1.58.5, from 10.1.58.5, 00:42:19 ago, via Ethernet0/1
      Route metric is 1536640, traffic share count is 4
      Total delay is 2002 microseconds, minimum bandwidth is 10000 Kbit
      Reliability 255/255, minimum MTU 1500 bytes
      Loading 1/255, Hops 2
R8#show eigrp address-family ipv6 topology 2001:db8:200::/48
EIGRP-IPv6 VR(AS200) Topology Entry for AS(200)/ID(8.8.8.8) for 2001:DB8:200::/48
  State is Passive, Query origin flag is 1, 2 Successor(s), FD is 196689920, RIB is 1536640
  Descriptor Blocks:
  FE80::A8BB:CCFF:FE00:600 (Ethernet0/1), from FE80::A8BB:CCFF:FE00:600, Send flag is 0x0
      Composite metric is (196689920/131153920), route is Internal
      Vector metric:
        Minimum bandwidth is 10000 Kbit
        Total delay is 2001250000 picoseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 2
        Originating router is 7.7.7.7
  FE80::A8BB:CCFF:FE00:700 (Ethernet0/2), from FE80::A8BB:CCFF:FE00:700, Send flag is 0x0
      Composite metric is (262225920/131153920), route is Internal
      Vector metric:
        Minimum bandwidth is 10000 Kbit
        Total delay is 3001250000 picoseconds
        Reliability is 255/255
        Load is 1/255
```